Lab 7

Fuzzing for Input Validation Bugs

L6_L7_42 - Salon Booking

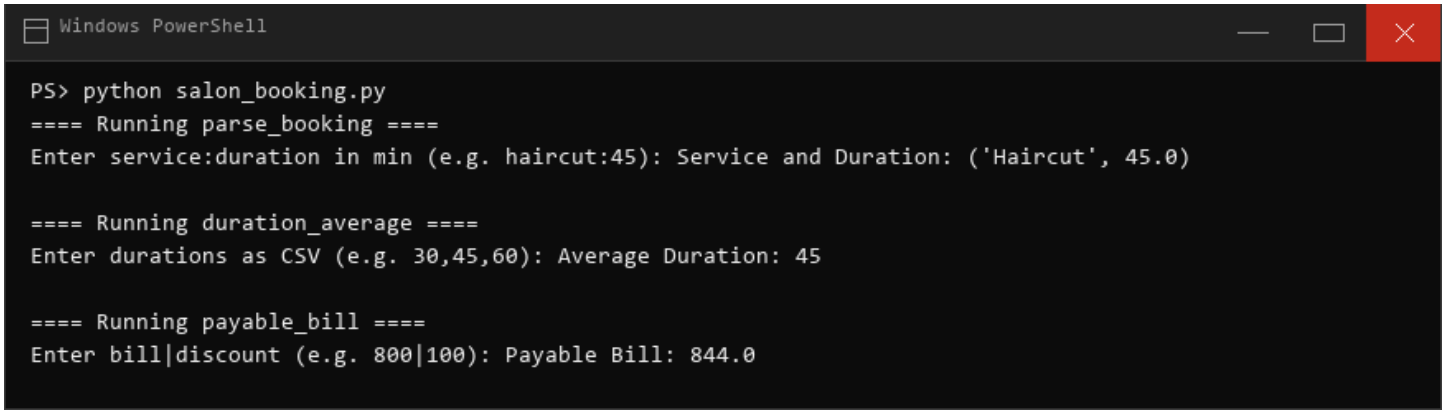
Tool: Hypothesis (property-based fuzz testing for Python)

Repository: github.com/akshat-k30/SE-Lab6-L6_L7_42_salon_booking

Step 2: Run the Buggy Application

Command: python salon_booking.py

The buggy code runs fine with well-formed inputs but has NO input validation. It will crash on None, empty strings, missing delimiters, or non-numeric values.



Step 4: Run Fuzz Tests Against Buggy Code - ALL FAILING

Command: pytest test_salon_booking.py -v
Hypothesis generates random inputs (including None). All 3 tests FAIL with AttributeError because the buggy functions call .split() on None.

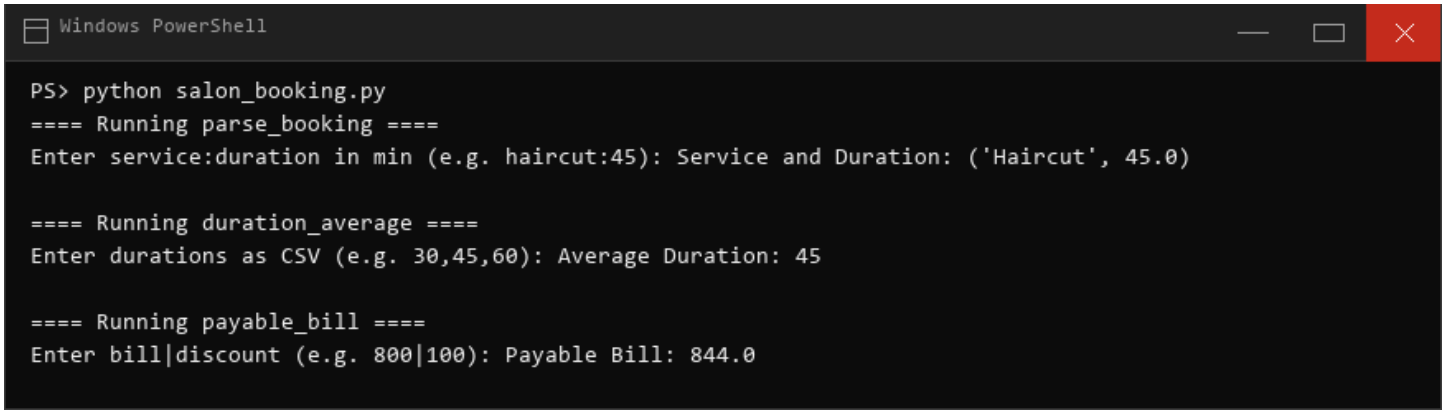
Note: The full pytest output is very long; the screenshot below shows the condensed key lines (header, FAILED markers, error types, and summary).



Step 5: Run the Fixed Application

Command: python salon_booking.py

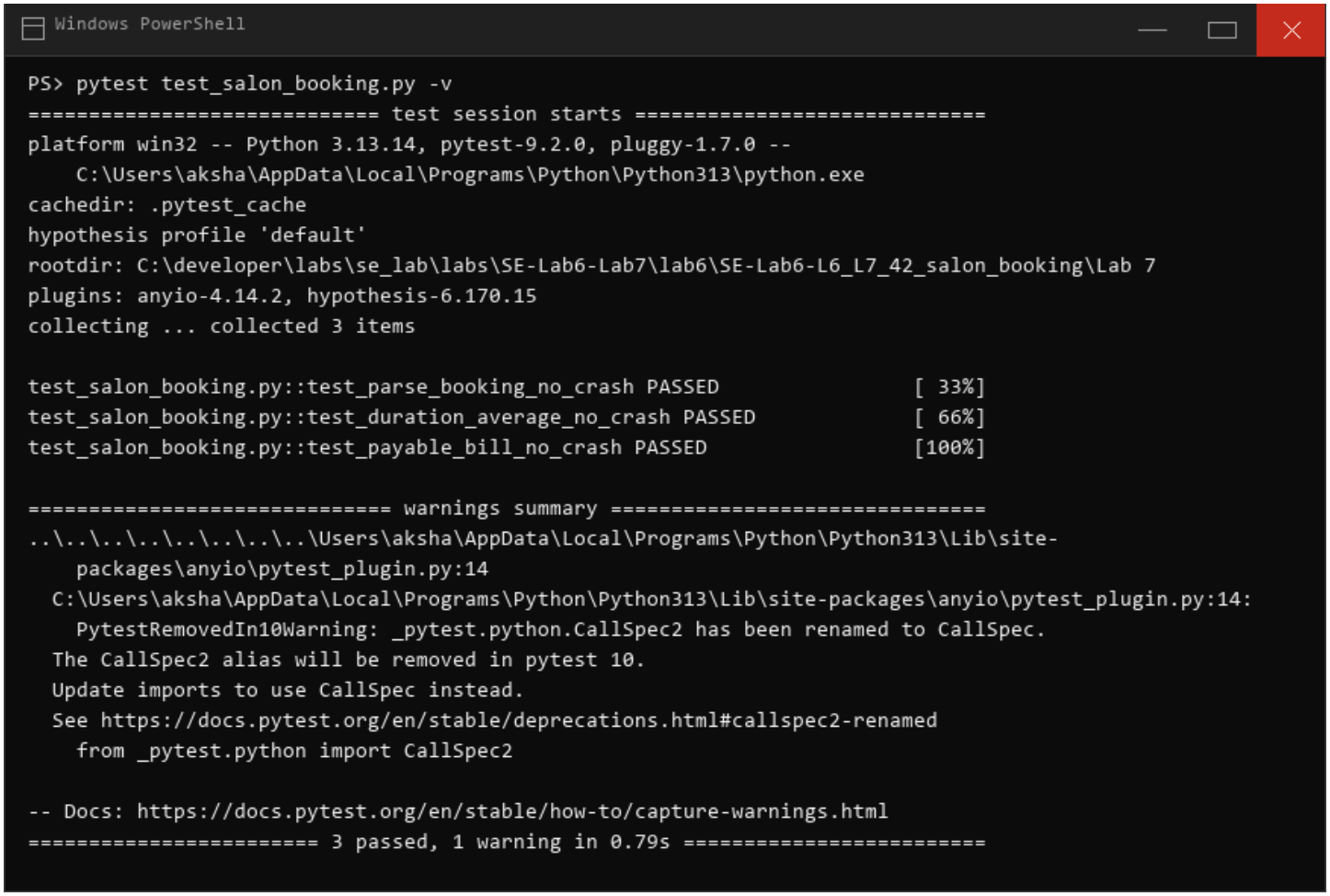
All three functions have been hardened with type checks, delimiter validation, empty-value detection, numeric parsing guards, and range validation. The fixed code produces the same correct output for valid inputs.



Step 6: Re-Run Fuzz Tests Against Fixed Code - ALL PASSING

Command: pytest test_salon_booking.py -v

All 3 Hypothesis fuzz tests now PASS. The fixed code correctly handles every edge case by raising ValueError instead of crashing.



Summary of Bugs Found and Fixes Applied

Bugs Discovered via Fuzz Testing:

1. parse_booking(None) --> AttributeError (no null check)
2. parse_booking('haircut') --> ValueError (missing ':' delimiter)
3. parse_booking('haircut:abc') --> ValueError (non-numeric duration)
4. parse_booking(':45') --> returns empty service name
5. duration_average(None) --> AttributeError (no null check)
6. duration_average('') --> ZeroDivisionError (empty list, div by zero)
7. duration_average('30,,60') --> ValueError (empty CSV element)
8. duration_average('30,abc,60') --> ValueError (non-integer value)
9. payable_bill(None) --> AttributeError (no null check)
10. payable_bill('800') --> ValueError (missing '|' delimiter)
11. payable_bill('abc|100') --> ValueError (non-numeric bill)

Fixes Applied (Defensive Programming):

- Type checking: reject None and non-string inputs with ValueError
- Delimiter validation: check for required separators (':', ',', '|')
- Empty value detection: reject empty fields after splitting
- Numeric parsing guards: try/except around int()/float() conversions
- Range validation: reject negative durations, negative bills
- Graceful error handling in main() with try/except for user input

Result:

All 3 Hypothesis fuzz tests pass. The functions now handle every edge case gracefully - returning correct results for valid inputs and raising ValueError with descriptive messages for invalid inputs. No unhandled exceptions remain.